

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO1: *Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems.(BL2,BL3)*

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Case Study

Duration: 2hrs

1. A logistics company is developing an RL-based warehouse robot that learns to pick, pack, and transport items efficiently while avoiding obstacles and minimizing delays. Analyze how the components of a Markov Decision Process (states, actions, rewards, policy, and environment) can be defined for this system. Design a reward function that balances speed, accuracy, and safety, and evaluate how reward design influences the robot's learning and behavior. **(5 Marks)**
2. A streaming platform wants to use Reinforcement Learning to recommend movies dynamically based on user preferences and viewing history. Apply RL concepts to construct a suitable state representation, action space, and reward function. Analyze how different reward signals affect user engagement and evaluate the trade-off between exploration and exploitation. **(5 Marks)**

3. An autonomous agricultural system uses RL to optimize irrigation and fertilizer usage based on soil conditions and weather forecasts. Design an RL framework by defining states, actions, and rewards. Analyze the effect of delayed rewards on crop yield optimization and justify the choice of a suitable learning algorithm.
(5 Marks)
4. A cybersecurity system uses RL to detect and respond to network attacks in real time. Analyze how states and environment can be modeled for intrusion detection. Design appropriate actions and reward functions, and evaluate the challenges of sparse rewards and real-time decision-making.
(5 Marks)
5. A ride-sharing company plans to use RL to optimize driver allocation and dynamic pricing strategies. Apply RL concepts to define states, actions, and rewards. Analyze how environmental uncertainty affects learning and evaluate ethical concerns related to pricing strategies.
(5 Marks)
6. A smart home automation system uses RL to control lighting, heating, and cooling based on occupancy and weather conditions. Design an RL framework including states, actions, and reward function. Analyze how conflicting objectives like comfort and energy saving can be handled and evaluate the impact of reward shaping.
(5 Marks)
7. A gaming company is developing an RL-based agent to play a complex strategy game against human players. Analyze how the problem can be modeled as an MDP. Design a reward structure that encourages long-term strategy and evaluate how exploration strategies affect performance.
(5 Marks)
8. A financial institution uses RL to detect fraudulent transactions in real time. Apply RL principles to define the state space, action space, and reward function. Analyze the challenges of imbalanced data and delayed feedback, and evaluate the risks associated with incorrect decisions.
(5 Marks)
9. A healthcare system uses RL to recommend personalized rehabilitation exercises for patients based on their progress. Design an RL framework with suitable states, actions, and rewards. Analyze the impact of delayed rewards and patient variability, and evaluate ethical concerns and safety considerations.
(5 Marks)

10. A smart grid system uses RL to manage electricity distribution and reduce peak load demand. Analyze how the components of an MDP can be defined in this context. Design a reward function to balance efficiency and reliability, and evaluate the challenges of scalability and real-time learning. **(5 Marks)**

Code of Conduct:

I M.Vineela (192472071) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Vineela

RUBRICS FOR CASESTUDY

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Question 1: RL-Based Warehouse Robot

Title

Autonomous Warehouse Robot Optimization Using Reinforcement Learning

Introduction

A logistics company manages a large warehouse where thousands of products need to be picked, packed, and transported daily. Manual operations may cause delays, incorrect product selection, and safety issues. To improve efficiency, the company develops an RL-based autonomous robot that learns through experience and makes intelligent decisions. The robot uses Reinforcement Learning (RL) where it interacts with the warehouse environment, receives rewards for good actions, and improves its decision-making process.

Problem Statement

The warehouse robot should:

- Pick the correct product
- Move efficiently inside the warehouse
- Avoid obstacles
- Reduce delivery time
- Maintain safety

MDP Components

1. State Representation (S)

The state represents the current situation of the robot.

It includes:

- Robot position
- Product location
- Battery level
- Distance from destination
- Obstacle information
- Current task status
- Number of pending orders

2. Action Space (A)

Actions are the possible decisions taken by the robot.

Examples:

- Move forward
- Move backward
- Turn left/right
- Pick product
- Drop product
- Pack item
- Deliver package

3. Environment (E)

The environment contains:

- Warehouse layout
- Storage shelves
- Products
- Workers
- Other robots
- Charging stations
- Obstacles

The robot learns by interacting with this environment.

4. Policy (π)

Policy defines how the robot chooses actions.

Example:

State	Action
Product found	Pick item
Obstacle detected	Change path
Battery low	Recharge
Delivery completed	Return

5. Reward Function

$$\text{Reward} = \text{Speed} + \text{Accuracy} + \text{Safety} - \text{Delay} - \text{Collision}$$

Situation	Reward
Correct item picked	+30
Successful delivery	+50
Fast movement	+20
Avoid obstacle	+10
Wrong item	-40
Collision	-100
Delay	-20

Reward Impact

- Speed-based reward only → Robot becomes fast but unsafe.
- Safety-based reward only → Robot becomes slow.
- Balanced reward → Robot becomes efficient and safe.

Conclusion

RL allows warehouse robots to learn optimal movement and task completion strategies. A proper reward function improves productivity, accuracy, and safety.

Question 2: RL-Based Movie Recommendation System

Title

Personalized Movie Recommendation Using Reinforcement Learning

Introduction

A streaming platform wants to recommend movies according to user interests. Traditional systems cannot quickly adapt to changing preferences. Reinforcement Learning helps the platform learn from user interactions and provide dynamic recommendations.

Problem Statement

The system must:

- Understand user preferences
- Recommend suitable movies
- Increase user engagement
- Improve viewing experience

RL Components

1. State Representation

The state represents user information:

- Watched movies
- Favourite genres
- Ratings
- Search history
- Viewing time
- User feedback

Example:

"User watches action movies and likes thriller content."

2. Action Space

Actions include:

- Recommend action movie
- Recommend comedy movie
- Recommend thriller movie
- Recommend new content

3. Environment

Environment includes:

- Users
- Movie database
- Streaming platform
- Feedback system

4. Reward Function

$$\text{Reward} = \text{User Satisfaction} + \text{Engagement}$$

User Response	Reward
Complete movie watched	+30
Like movie	+15
Add to watchlist	+10
Skip movie	-10
Stop early	-20

Exploration vs Exploitation

Exploration

The system recommends new movies.

Advantages:

- Discovers hidden interests
- Provides variety

Exploitation

The system recommends previously liked movies.

Advantages:

- Higher immediate satisfaction

Reward Effect

Poor reward:

- Only watch time → Long movies preferred.

Balanced reward:

- Improves satisfaction and recommendation accuracy.

Conclusion

RL helps streaming platforms provide personalized recommendations by continuously learning user behaviour.

Question 3: RL-Based Smart Home Automation

Title

Energy Efficient Smart Home Control Using Reinforcement Learning

Introduction

A smart home system uses RL to automatically control lighting, heating, and cooling. The goal is to reduce electricity usage while maintaining user comfort.

Problem Statement

The system should:

- Save energy
- Maintain comfortable temperature
- Automatically control appliances

RL Framework

1. State Representation

State includes:

- Room temperature
- Occupancy level
- Weather condition

- Time of day
- Energy consumption

2. Action Space

Actions:

- Turn lights ON/OFF
- Increase/decrease AC temperature
- Control heater
- Open ventilation

3. Environment

Includes:

- Home appliances
- Residents
- Weather conditions
- Electricity supply

4. Reward Function

$$\textit{Reward} = \textit{Comfort} + \textit{EnergySaving} - \textit{Cost}$$

Condition	Reward
Comfortable temperature	+30
Energy saved	+20
Correct lighting	+10
Energy waste	-20
User discomfort	-30

Conclusion

RL helps smart homes automatically manage energy consumption while maintaining user comfort.

Question 4: RL-Based Strategy Game Agent

Title

Artificial Intelligence Game Player Using Reinforcement Learning

Introduction

A gaming company develops an RL agent to play complex strategy games against humans. The agent must learn planning, resource management, and long-term strategies.

Problem Statement

The agent should:

- Make strategic decisions
- Defeat opponents
- Manage resources
- Improve gameplay

MDP Components

1. State

Includes:

- Player health
- Enemy position

- Available resources
- Unit locations
- Game level

2. Actions

Actions:

- Attack enemy
- Defend base
- Collect resources
- Upgrade weapons
- Build structures

3. Environment

Includes:

- Game map
- Opponents
- Resources
- Obstacles

4. Reward Function

$$\textit{Reward} = \textit{Winning Benefit} - \textit{Loss Penalty}$$

Event	Reward
Win game	+100
Destroy enemy	+20
Collect resources	+10
Lose unit	-20
Lose game	-100

Exploration and Exploitation

Exploration:

- Tests new strategies

Exploitation:

- Uses successful strategies

Balanced learning improves performance.

Conclusion

RL allows game agents to develop advanced strategies and compete effectively with human players.

Question 5: RL-Based Smart Grid Management

Title

Intelligent Electricity Distribution Using Reinforcement Learning

Introduction

A smart grid system uses RL to manage electricity distribution efficiently. It aims to reduce peak load, minimize energy waste, and maintain reliable power supply.

Problem Statement

The system must:

- Balance electricity demand and supply
- Reduce energy wastage
- Improve reliability

MDP Components

1. State Representation

Includes:

- Electricity demand
- Power generation
- Battery storage

- Weather conditions
- Time period

2. Action Space

Actions:

- Increase power supply
- Reduce load
- Charge battery
- Discharge battery
- Shift energy usage

3. Environment

Contains:

- Power stations
- Consumers
- Renewable sources
- Storage systems

4. Reward Function

$$\textit{Reward} = \textit{Efficiency} + \textit{Reliability} - \textit{EnergyLoss}$$

Situation	Reward
Meet electricity demand	+30
Reduce peak load	+20
Efficient energy usage	+15
Power failure	-50
Energy waste	-20

Challenges

Scalability

- Large number of users
- Huge state space
- High computational requirements

Real-Time Learning

- Demand changes continuously
- Weather uncertainty
- Fast decisions required

Conclusion

Reinforcement Learning provides an intelligent solution for smart grid management by optimizing energy distribution, reducing wastage, and maintaining reliability.